COSMIC CALCULATION SERIES / ORIGINAL PRACTICE

Concept of Time

02.02 · Original club practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

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These are original MAIS PAC practice exercises, not official IOAA problems.

PROBLEMS

01 Solar and sidereal days

A mean solar day is 24 h and a sidereal day is approximately 23 h 56 min 4 s. (a) Explain why the sidereal day is shorter in terms of Earth’s orbit. (b) A star transits at 22:00 mean solar time tonight. Estimate its transit time 30 days later, ignoring small variations. (c) Explain why the result is approximate.

02 Longitude and local time

Two observers are at longitudes 105° E and 75° E. Ignore the equation of time. (a) Find the difference between their local mean solar times. (b) If it is 12:00 local mean solar time at 105° E, what is it at 75° E? (c) Explain why civil time zones need not agree exactly with local mean solar time.

03 Julian dates

Julian dates change continuously in days, and JD 2451545.0 corresponds to 2000 January 1 at 12:00 UT. (a) What is the JD at 18:00 UT on the same date? (b) What is the JD at 00:00 UT on January 2? (c) If an event at JD 2451545.25 lasts 90 minutes, find the JD at its end. Ignore leap seconds.

04 Different years and timing corrections

A sidereal year measures Earth’s orbital period relative to distant stars; a tropical year measures the cycle of seasons. (a) Explain why these definitions differ because of precession. (b) Define the equation of time as apparent solar time minus mean solar time, and explain qualitatively why it varies through the year. (c) Why can a heliocentric Julian date be useful when comparing observations of a periodic astronomical event taken months apart?